



Introduction to Photogrammetry

Lecture 13

Introduction to Analytical Photogrammetry

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In Today's Class

1. Introduction
2. Concept of Coordinate Systems

1. Introduction

- Photogrammetry is the science and technology of obtaining reliable measurements from photographs.
- It is used to determine the size, shape, position, and elevation of objects on the Earth's surface.
- Photogrammetry converts two-dimensional image information into three-dimensional spatial data.
- The technique is widely used in mapping, surveying, engineering, and remote sensing.

- Early photogrammetry was performed using analog instruments and manual measurements.
- Analytical photogrammetry introduced mathematical modeling and computer-based calculations.
- Modern digital photogrammetry uses digital images and automated processing techniques.
- The development of computers greatly improved the accuracy and efficiency of photogrammetric analysis.

Importance of Analytical Photogrammetry

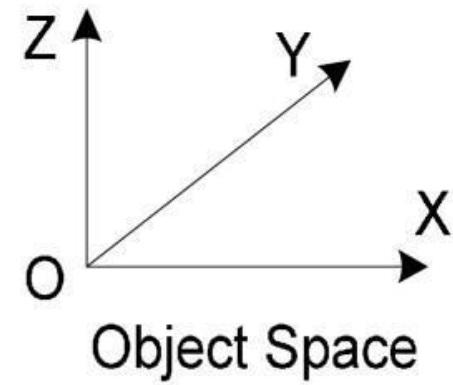
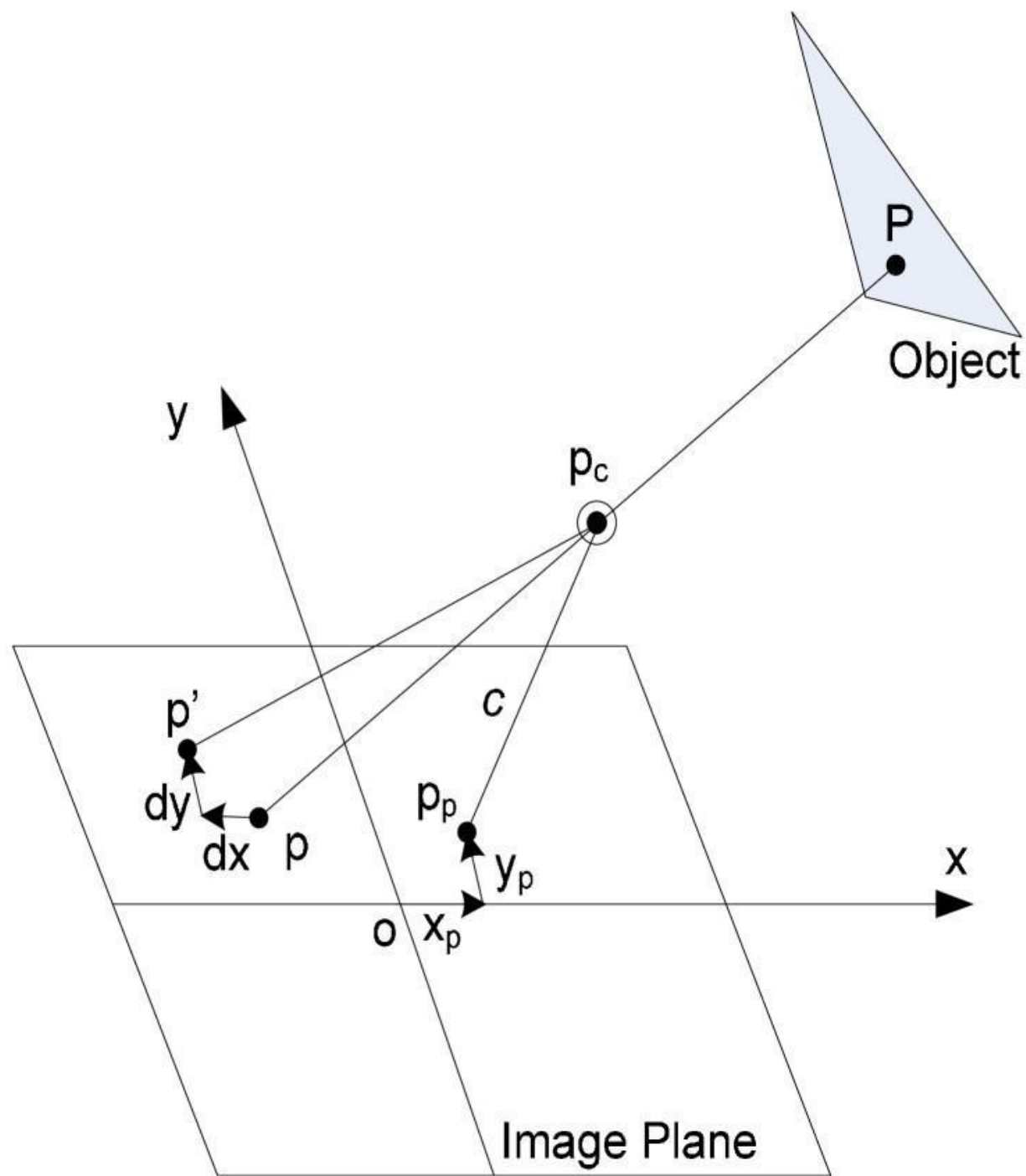
- **Analytical photogrammetry** provides highly accurate spatial measurements from aerial photographs.
- It allows the reconstruction of real-world geometry using mathematical equations.
- The method is essential for topographic mapping and terrain modeling.
- It supports applications in urban planning, disaster management, and environmental monitoring.

Applications of Analytical Photogrammetry

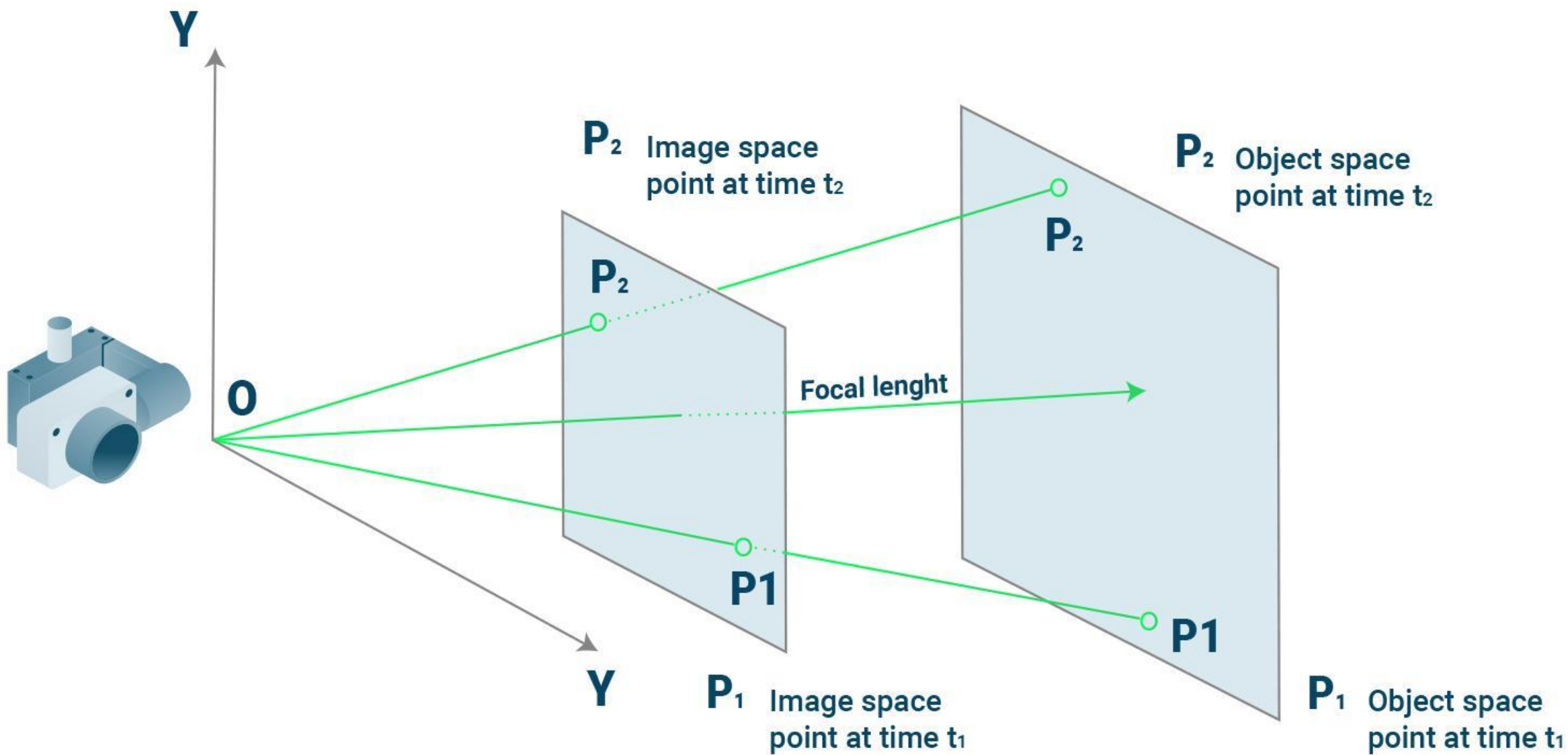
- Analytical photogrammetry is used in the production of topographic maps.
- It is widely applied in transportation engineering and infrastructure development.
- The technique helps in monitoring land use and land cover changes over time.
- It is also used in archaeology, forestry, and military intelligence.

Basic Principle of Photogrammetry

- Photogrammetry is based on the principle of central perspective projection.
- Light rays travel from the ground object through the camera lens onto the image plane.
- Every image point corresponds to a point on the ground surface.
- Mathematical relationships are used to reconstruct object positions from image measurements.



p_c - perspective center
 P - object point
 p' - perturbed image point
 p - ideal image point
 p_p - principal point
 c - principal distance



Components of a Photogrammetric System

- A photogrammetric system consists of a camera, photographs, and measurement instruments.
- The camera captures overlapping aerial images of the terrain.
- Control points are used to relate image coordinates to ground coordinates.
- Computer software performs calculations and spatial reconstruction.

Difference Between Analog and Analytical Photogrammetry

- Analog photogrammetry relies mainly on optical and mechanical instruments.
- Analytical photogrammetry uses mathematical models and digital computations.
- Analytical methods provide greater flexibility and higher accuracy.
- Computer-based processing allows automation of many photogrammetric tasks.

2. Concept of Coordinate Systems

- Coordinate systems are essential for locating points in space accurately.
- Different coordinate systems are used for image measurements and ground measurements.
- Photogrammetry requires transformations between multiple coordinate systems.
- Accurate coordinate handling is necessary for reliable spatial reconstruction.

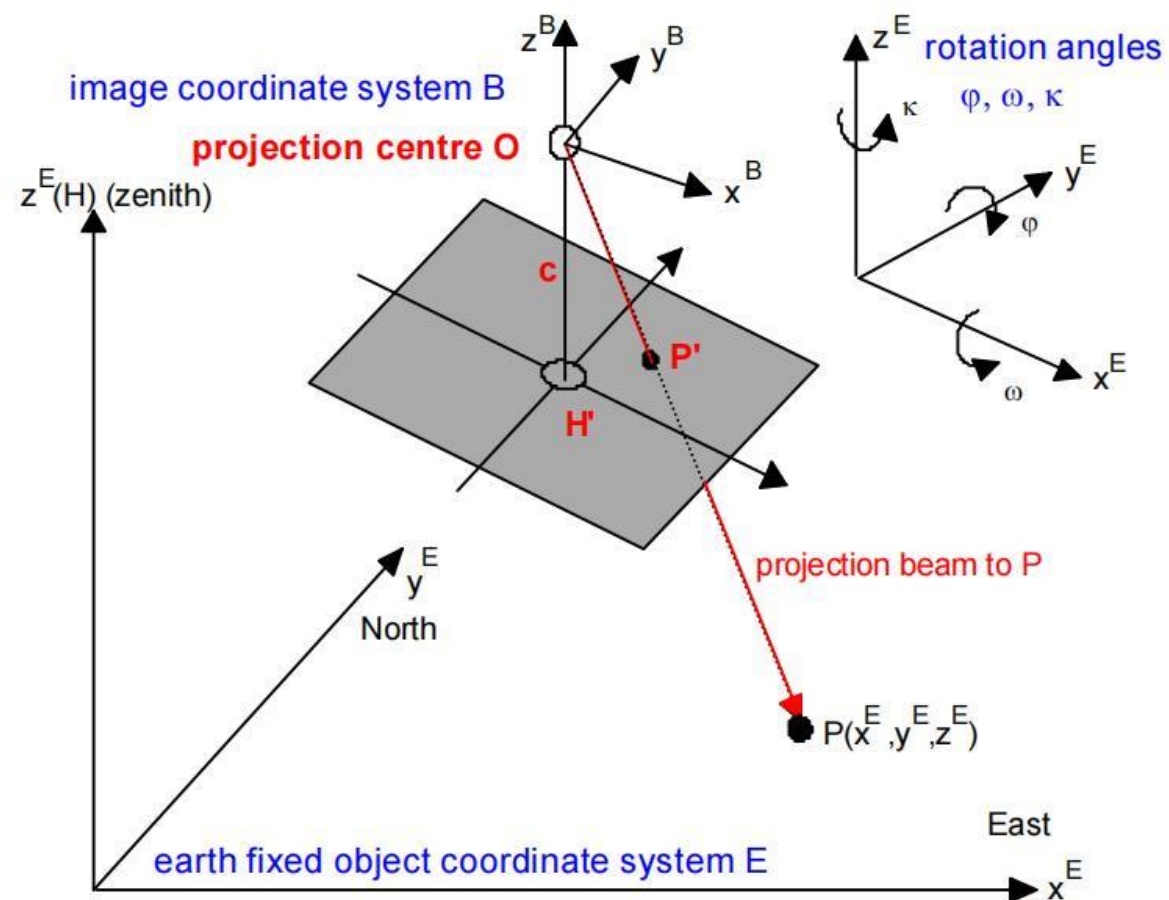
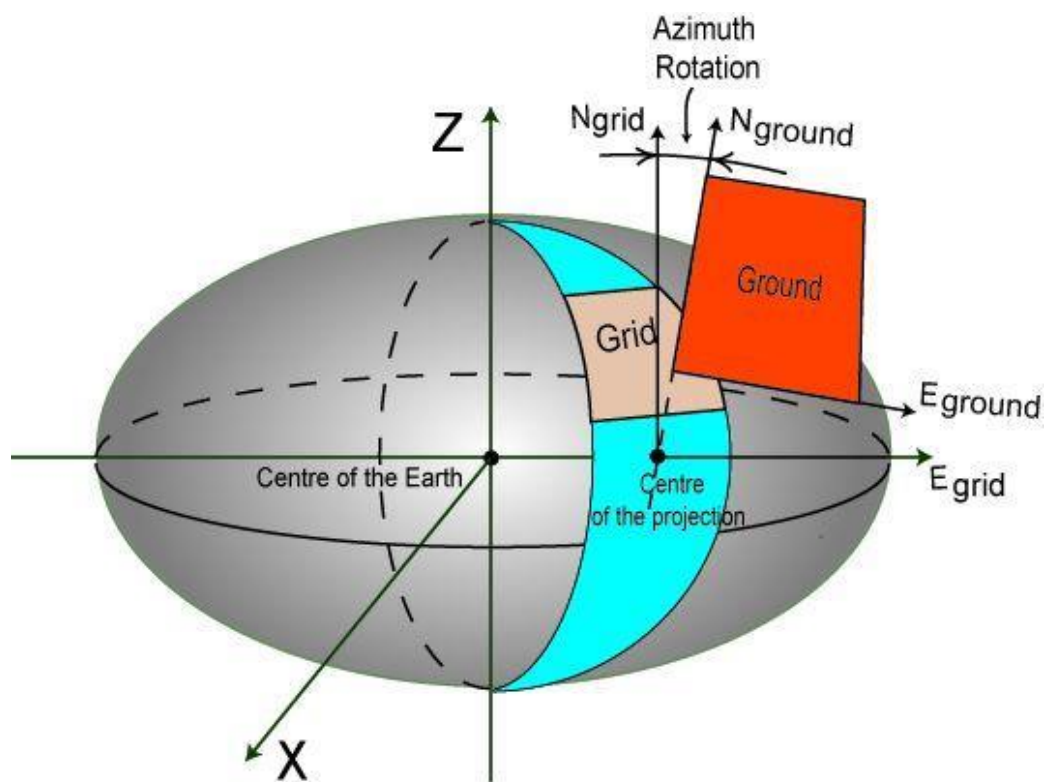


Image Coordinate System

- The image coordinate system is defined on the photograph or image plane.
- The origin is usually located at the principal point of the photograph.
- Coordinates are expressed in terms of x and y distances on the image.
- Image coordinates are used during measurement and orientation processes.

Fiducial Coordinate System

- Fiducial marks are reference marks placed at the edges of aerial photographs.
- The fiducial coordinate system provides a stable reference framework for measurements.
- Interior orientation uses fiducial marks to reconstruct the camera geometry.
- Accurate fiducial measurements improve the precision of photogrammetric analysis.

Fiducial Coordinate System

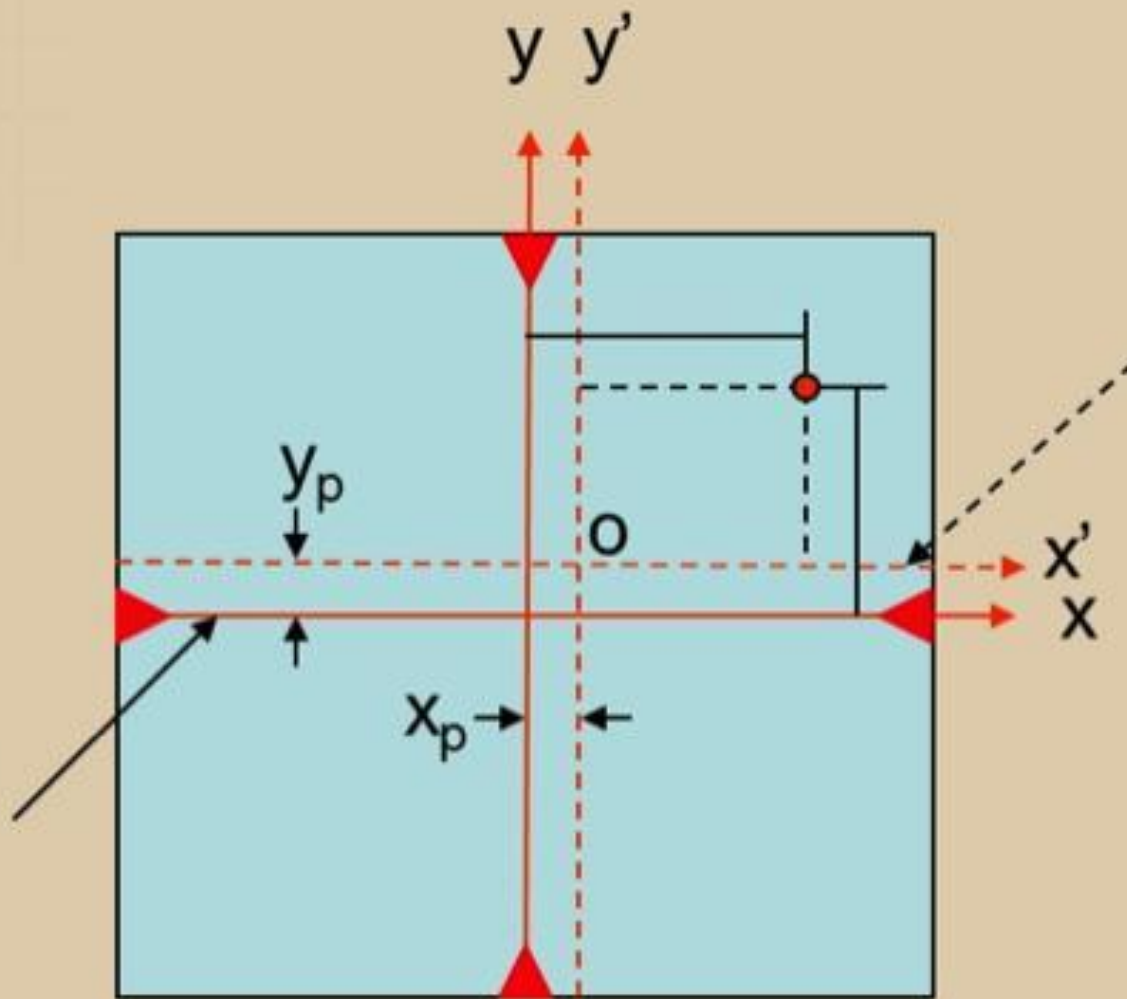


Photo Coordinate System

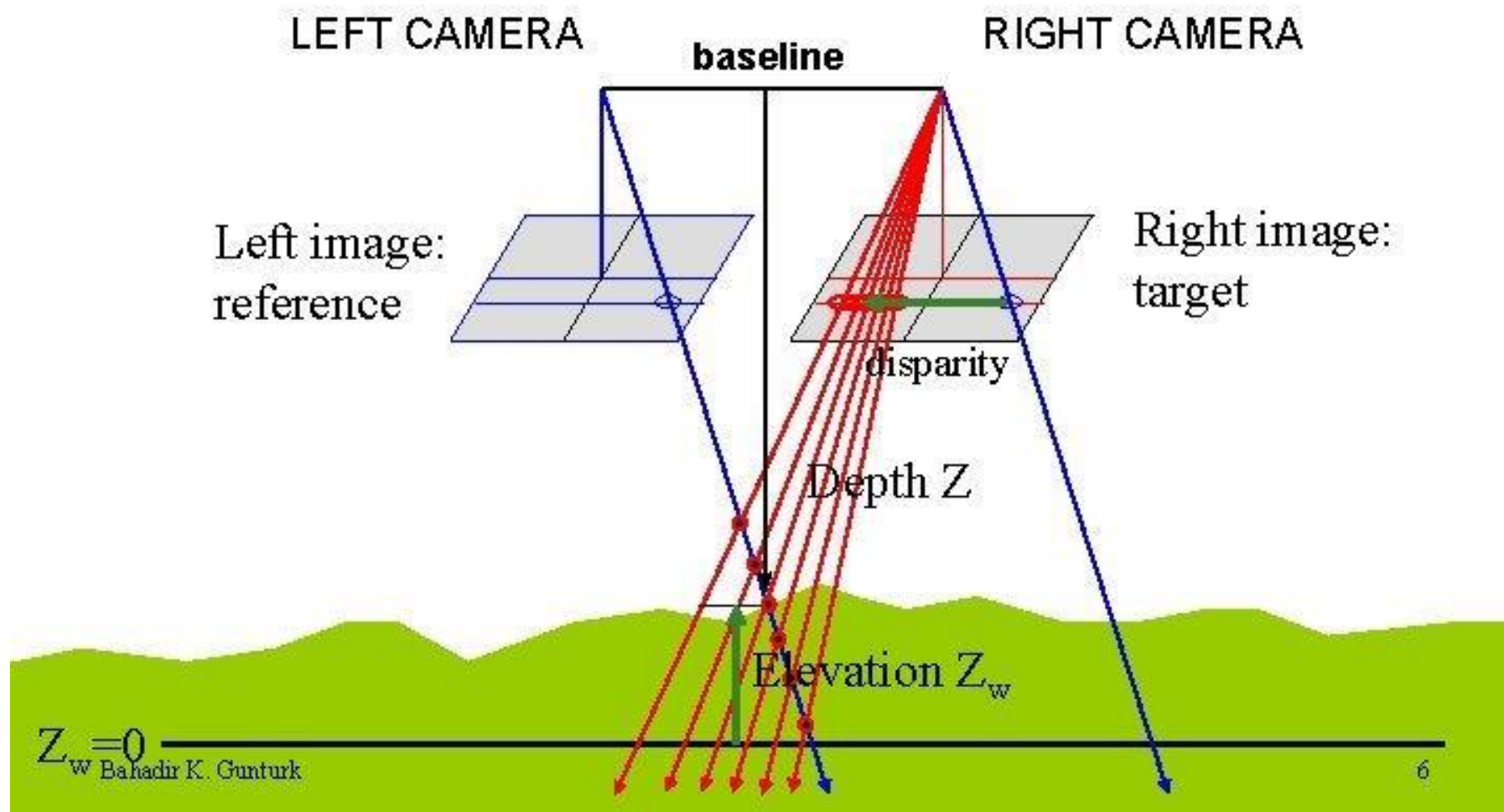
Ground Coordinate System

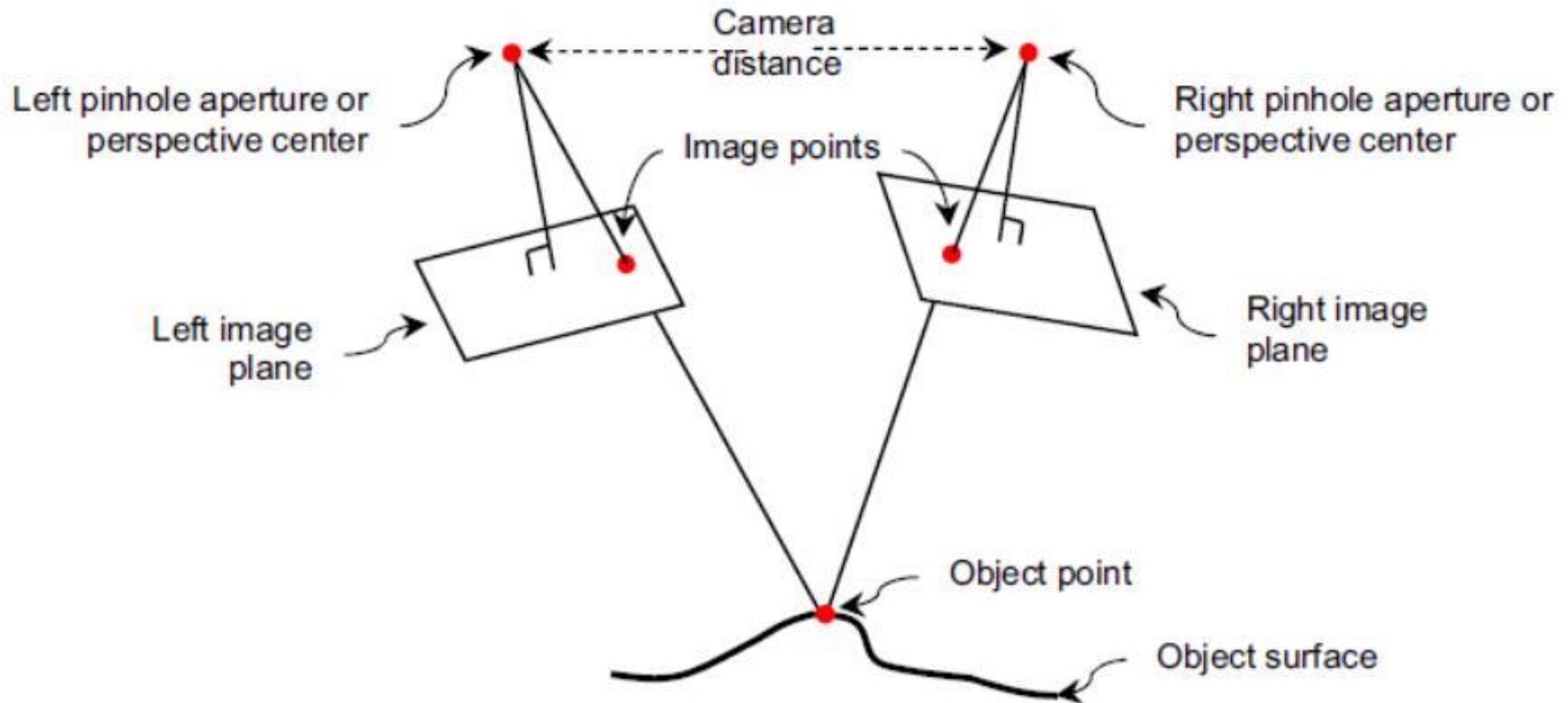
- The ground coordinate system represents the actual positions of objects on the Earth.
- Ground coordinates are commonly expressed as X, Y, and Z values.
- X and Y represent horizontal positions, while Z represents elevation.
- Ground control points connect image space with real-world space.

Model Coordinate System

- The model coordinate system is created during stereo image reconstruction.
- It represents a three-dimensional model formed from overlapping photographs.
- Relative orientation establishes the geometry of the stereo model.
- The model system must later be transformed into the ground coordinate system.

A Simple Stereo System







Thank you!

Any Questions
